



Sri Lanka Institute of Information Technology

Data Warehousing and Business Intelligence IT3021

Assignment 2 2025

Group : Year 3 Semester 2 DS WE 1.1
Malabe Campus

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Step 1: Data source for the Assignment 2

1.1 Data Source Overview

The data source for Assignment 2 is the **Healthcare_DW** database implemented in Assignment 1, hosted on SQL Server. It is built on synthetic healthcare records from the Synthea COVID-19 dataset representing the fictional hospital network "**Unity Health Group**", covering the period 2019–2020 with over 321,528 encounter transactions.

The warehouse follows a Snowflake Schema. The Provider dimension is normalized through the Organization dimension via OrgKey rather than connecting directly to the fact table, creating the snowflake effect.

Fact Table — FactEncounter

The central fact table stores 321,528 healthcare transaction records with the following measures:

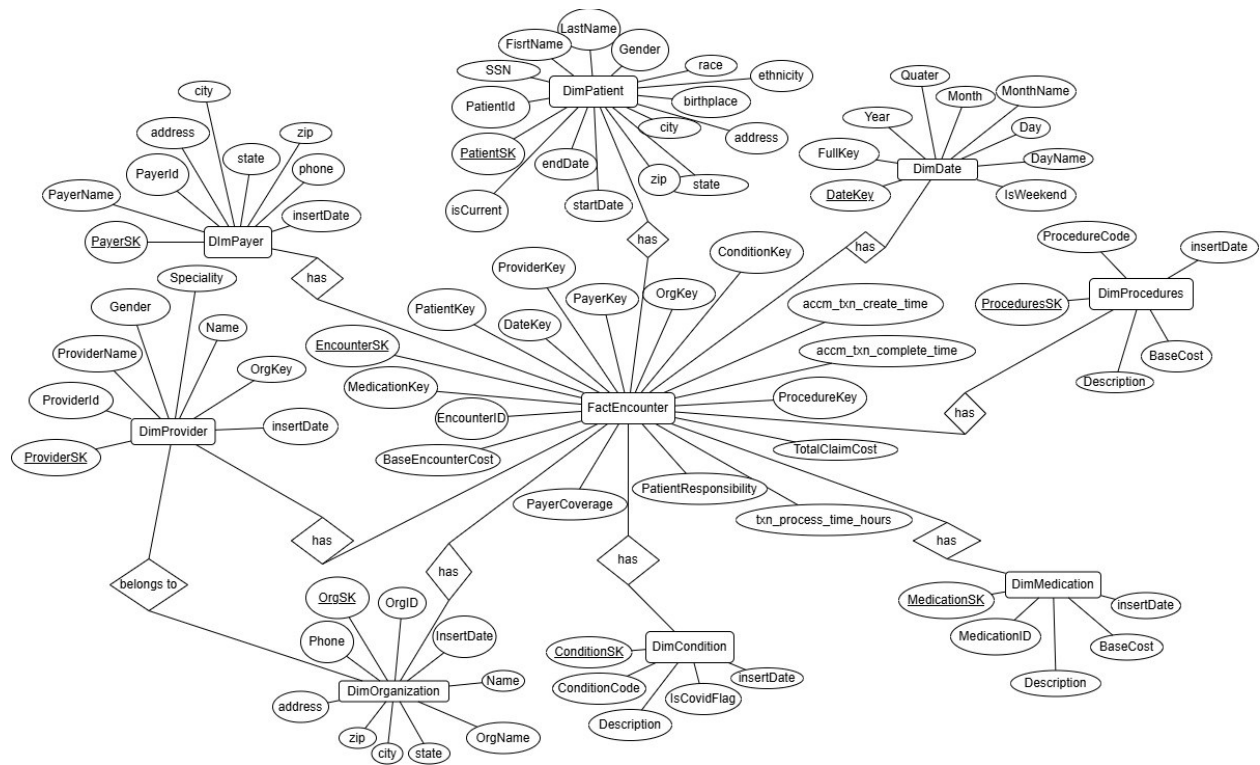
- BaseEncounterCost - Base cost of the hospital visit
- TotalClaimCost - Total amount claimed from insurance
- PayerCoverage - Amount the payer covered
- PatientResponsibility - Uncovered cost borne by the patient (computed)
- accm_txn_create_time - Timestamp when the record was first loaded
- accm_txn_complete_time - Timestamp when the transaction was completed
- txn_process_time_hours - Hours between creation and completion

Dimension Tables

Dimension	Surrogate Key	Key Attributes
DimDate	DateKey	Year, Quarter, Month, Day, DayName, IsWeekend
DimPatient	PatientSK	FirstName, LastName, Gender, Race, Address, City, State — SCD Type 2
DimPayer	PayerSK	PayerName, Address, City, State, Phone
DimProvider	ProviderSK	ProviderName, Gender, Speciality, OrgKey
DimOrganization	OrgSK	OrgName, Address, City, State, Phone
DimCondition	ConditionSK	ConditionCode, Description, IsCovidFlag
DimMedication	MedicationSK	MedicationID, Description, BaseCost
DimProcedures	ProcedureSK	ProcedureCode, Description, BaseCost

Slowly Changing Dimension

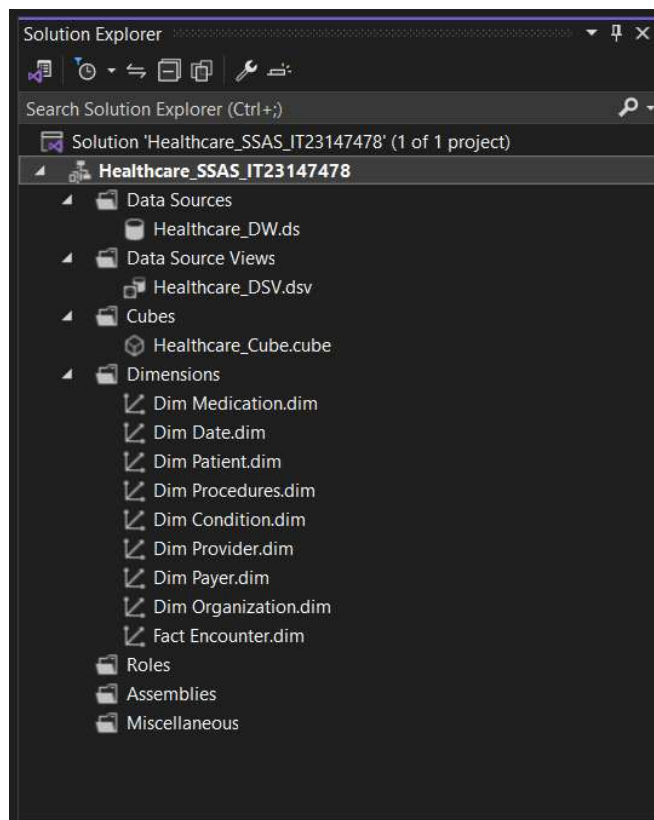
DimPatient is implemented as **SCD Type 2** — changes to a patient's Address trigger a new record with StartDate, EndDate, and IsCurrent flag to preserve historical context. LastName is handled as SCD Type 1 (overwrite).



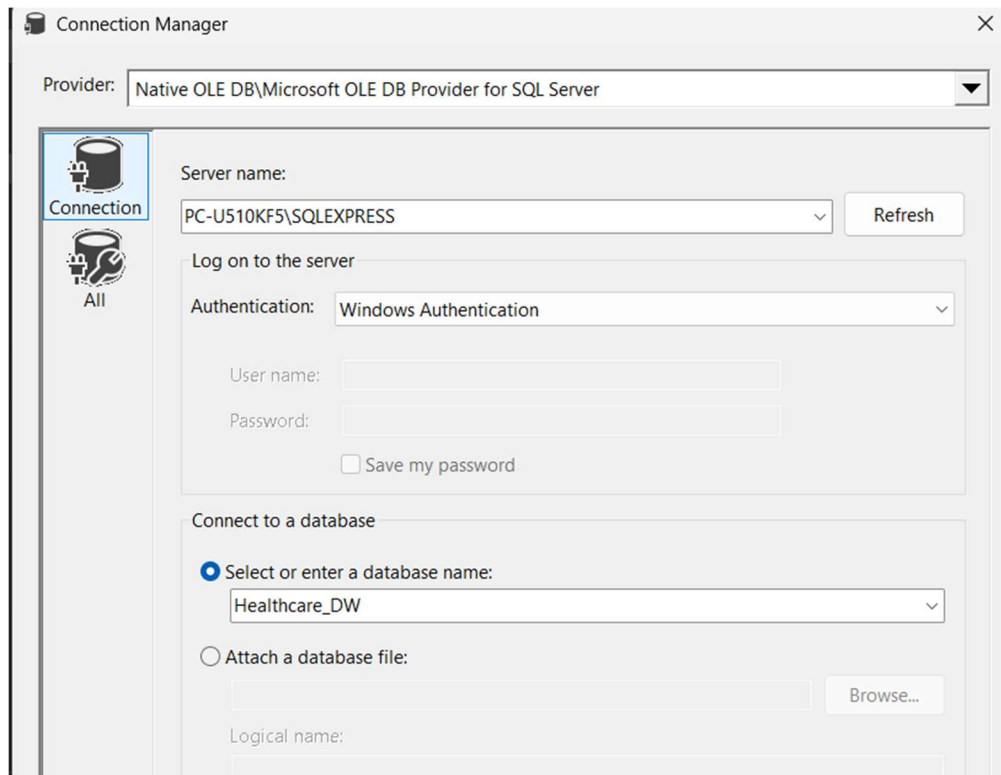
Step 2: SSAS Cube Implementation

2.1 Creating the SSAS Project

A new Analysis Services Multidimensional project named Healthcare_SSAS_IT23147478 was created in Visual Studio (SSDT). This project serves as the analytical layer on top of the Healthcare_DW data warehouse.



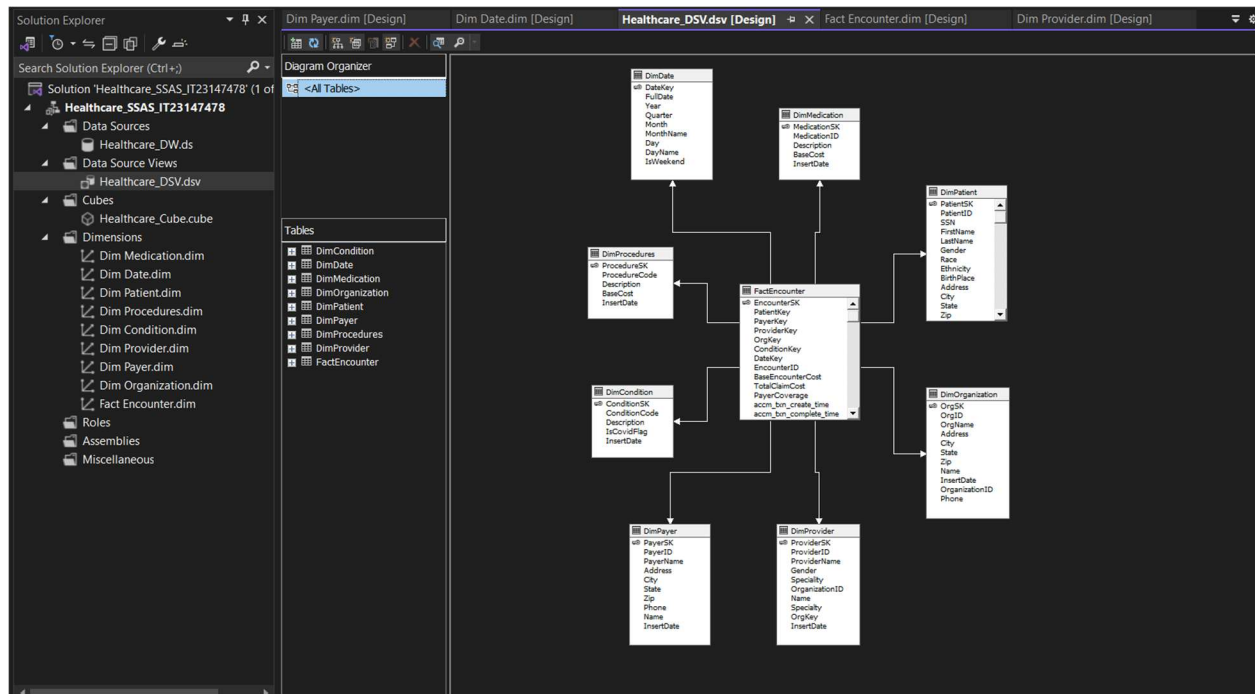
2.2 Data Source Configuration



A data source named Healthcare_DW was configured to connect to the SQL Server instance at PC-U510KF5\SQLEXPRESS using the Microsoft OLE DB Provider for SQL Server. The initial catalog was set to Healthcare_DW. The impersonation setting was configured to use the service account (NT Service\MSOLAP\$SSAS) which was granted db_datareader and db_owner permissions on the Healthcare_DW database.

2.3 Data Source View

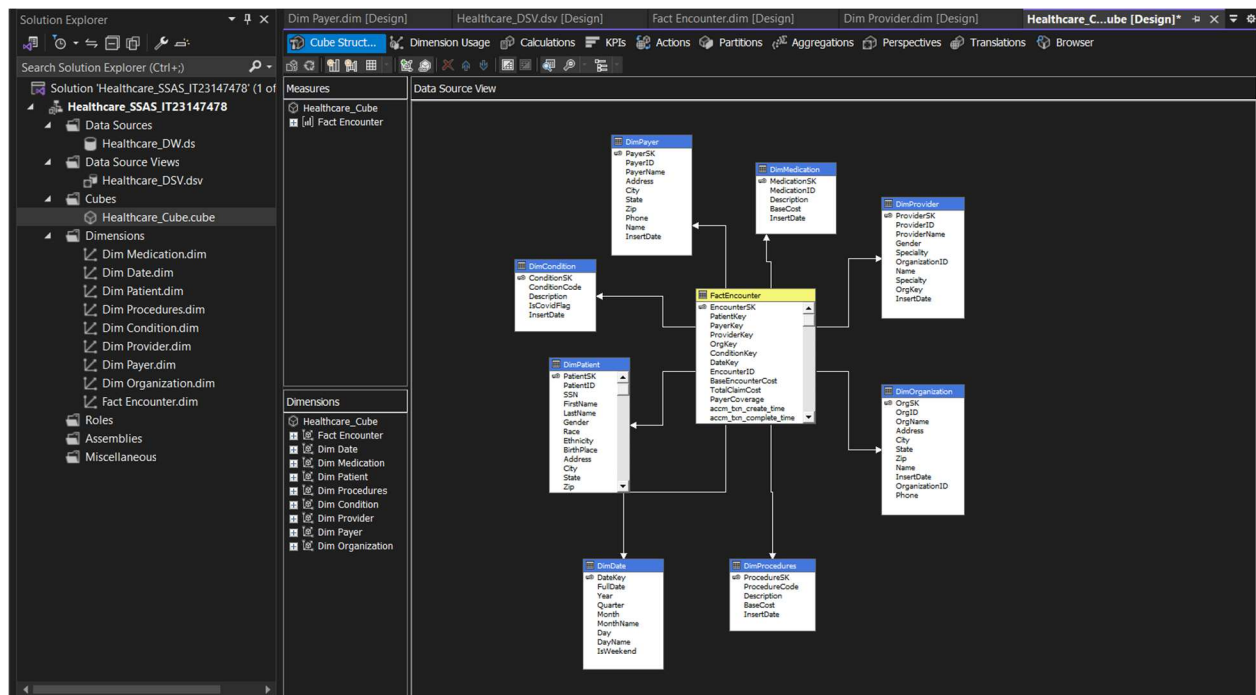
A Data Source View named Healthcare_DSV was created including all 9 tables from the data warehouse — FactEncounter and all 8 dimension tables. SSAS automatically detected the foreign key relationships defined in the database, correctly linking all dimension surrogate keys to the fact table.



2.4 Cube Design

The cube named Healthcare_Cube was designed using the Cube Wizard with the following configuration:

- Fact table: FactEncounter
- Measures included:
 - Base Encounter Cost - Base cost of each hospital visit
 - Total Claim Cost - Total insurance claim amount
 - Payer Coverage - Amount covered by insurance
 - Patient Responsibility - Uncovered cost borne by patient
- Dimensions connected: Dim Date, Dim Patient, Dim Payer, Dim Provider, Dim Organization, Dim Condition, Dim Medication, Dim Procedures



2.5 Hierarchy Implementation

Two hierarchies were implemented:

Calendar Hierarchy (in Dim Date):

Year → Quarter → Month → Day

Attribute relationships were defined as: Day relates to Month, Month relates to Quarter, Quarter relates to Year. This enables time-based drill-down analysis from yearly trends down to daily granularity.

Location Hierarchy (in Dim Organization):

State → City → Org Name

This enables geographic drill-down analysis of healthcare costs and encounter volumes across the hospital network.

Attributes	Hierarchies
 Dim Date - Date Key - Day - Day Name - Full Date - Is Weekend - Month - Month Name - Quarter - Year	Calendar Hierarchy ☒ - Year - Quarter - Month - Day <new level> To create a new hierarchy, drag an attribute here.
Attributes	Hierarchies
 Dim Organization - Address - City - Insert Date - Name - Org ID - Org Name - Org SK - Phone - State - Zip	Location Hierarchy ☒ - State - City - Org Name <new level> To create a new hierarchy, drag an attribute here.

2.6 Deployment and Processing

The project was deployed to the Analysis Services instance at PC-U510KF5\SSAS. The cube was processed successfully with **321,528 fact rows** read from FactEncounter, along with all dimension attributes processed correctly.

Deployment Progress - Healthcare_SSAS_I...
Server : PC-U510KF5\SSAS
Database : Healthcare_SSAS_IT23147478
Command
Status:

Deployment Completed Successfully

es
packa
ge
docum
ent.

Dimensions

Dim Medication.dim

Dim Date.dim

Dim Patient.dim

Dim Procedures.dim

Dim Condition.dim

Dim Provider.dim

Dim Payer.dim

Dim Organization.dim

Fact Encounter.dim

Roles

Assemblies

Miscellaneous

Process Progress

Command

Processing Database 'Healthcare_SSAS_IT23147478' completed.

Processing Dimension 'Dim Condition' completed.

Processing Dimension 'Dim Date' completed.

Processing Dimension 'Dim Medication' completed.

Processing Dimension 'Dim Organization' completed.

Processing Dimension 'Dim Patient' completed.

Processing Dimension 'Dim Payer' completed.

Processing Dimension 'Dim Procedures' completed.

Processing Dimension 'Dim Provider' completed.

Processing Dimension 'Fact Encounter' completed.

Processing Cube 'Healthcare_Cube' completed.

Start time: 4/3/2026 7:13:33 PM; End time: 4/3/2026 7:13:43 PM; Duration: 0:00:09

Processing Measure Group 'Fact Encounter' completed.

Start time: 4/3/2026 7:13:33 PM; End time: 4/3/2026 7:13:43 PM; Duration: 0:00:09

Status:

Process succeeded.

Stop

Reprocess

View Details...

Copy

Close

Help

Step 3: Demonstration of OLAP operations

3.1 Connecting Excel to the SSAS Cube

Excel was connected to the SSAS cube using the Data Connection Wizard. In Excel, navigating to the Data tab and selecting Get Data → From Other Sources → From Analysis Services opened the Data Connection Wizard. The server name PC-U510KF5\SSAS was entered and the Healthcare_SSAS_IT23147478 database was selected along with the Healthcare_Cube. The connection was saved as an ODC file and the data was imported as a PivotTable Report on a new worksheet. The PivotTable Fields panel on the right displays all available measures from FactEncounter and all connected dimensions.

3.2 Roll-Up

Roll-up aggregates data from a lower level of detail to a higher level, summarizing measures across a dimension hierarchy.

The Calendar Hierarchy from Dim Date was dragged to the Rows area and Total Claim Cost was added to the Values area. The PivotTable initially displayed data at the finest granularity — individual years within the hierarchy. The data was then collapsed upward to display Total Claim Cost aggregated by Year, showing:

- 2019: 3,277,407.04
- 2020: 3,341,264.64
- Grand Total: 41,384,265.92

This demonstrates roll-up — data from individual daily and monthly records has been aggregated upward to the Year level and finally to a single Grand Total, reducing the level of detail while increasing the scope of analysis.

Row Labels	Total Claim Cost
1910	516.64
1911	1162.44
1912	2324.88
1913	1549.92
1914	1679.08
1915	4158.92
1916	4133.12
1917	3564.8
1918	3435.64
1919	6587.16
1920	4081.44
1921	4778.92
1922	5321.36
1923	5295.56
1924	5450.52
1925	5631.36
1926	5553.88
1927	7594.56
1928	9377
1929	10875.24
1930	13716.76
1931	18289.04
1932	23765.44
1933	24747.04
1934	32238.32
1935	35751.44
1936	34149.84
1937	36423.12

Roll-up: Total Claim Cost by Year

105	2013	957307.4	
106	2014	960200.36	
107	2015	956738.92	
108	2016	998845.16	
109	2017	1000291.72	
110	2018	1123794.48	
111	2019	3277407.04	
112	2020	3341264.64	
113	Grand Total	41384265.92	
114			

Roll-up: Grand Total

3.3 Drill-Down

Drill-down is the opposite of roll-up — navigating from a higher level of aggregation to a lower, more detailed level.

Starting from the Year level view, the + icon next to 2018 was clicked to expand to the Quarter level, revealing quarterly totals (Q1, Q2, Q3, Q4). The + next to Quarter 1 was then clicked to expand to the Month level, showing individual month totals. Finally expanding a month revealed individual Day level data showing the most granular view of Total Claim Cost per day.

This demonstrates the full drill-down path through the Calendar Hierarchy:

Year → Quarter → Month → Day

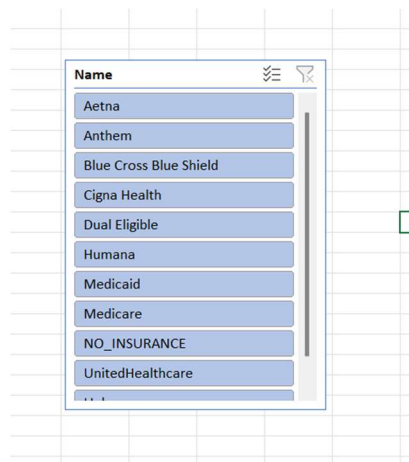
Row Labels	Total Claim Cost
2018	
1	
1	
1	3874.8
10	2583.2
11	2841.52
12	2841.52
13	2583.2
14	3229
15	4262.28
16	2919
17	2970.68
18	2712.36
19	3513.12
2	2324.88
20	2660.68
21	1808.24
22	2712.36
23	3229
24	1885.72
25	2712.36
26	2841.52
27	2970.68
28	3745.64
29	3229
3	2660.68
30	2066.56
31	2583.2
4	2402.36
5	2712.36
6	2712.36
7	3099.84
8	2789.84
9	4133.12
2	75739.36
3	85452.16
2	
4	80802.4
5	87286.28
6	90127.8
3	

3.4 Slice

Slicing filters the data cube by selecting a specific value from one dimension, effectively cutting a single "slice" through the data.

With the PivotTable showing Total Claim Cost by Year, a Slicer was inserted via PivotTable Analyze → Insert Slicer → selecting Name from Dim Payer. The slicer displayed all available payers: Aetna, Anthem, Blue Cross Blue Shield, Cigna Health, Dual Eligible, Humana, Medicaid, Medicare, NO_INSURANCE, and UnitedHealthcare.

Before filter: The PivotTable showed Total Claim Cost across all payers for all years.



After filter: Clicking **Medicaid** in the slicer filtered the entire PivotTable to show only Medicaid's claim costs across years. This demonstrates slicing — the data cube was cut along the Payer dimension to show only one payer's data.

	A	B	C	D	E	F	G	H	I	J	K
1	Row Labels	Total Claim Cost									
2	1912	258.32									
3	1914	258.32									
4	1915	258.32									
5	1916	1033.28									
6	1917	774.96									
7	1918	904.12									
8	1919	645.8									
9	1920	516.64									
10	1921	645.8									
11	1922	387.48									
12	1923	387.48									
13	1924	1162.44									
14	1925	723.28									
15	1926	387.48									
16	1927	1162.44									
17	1928	516.64									
18	1929	645.8									
19	1930	1575.72									
20	1931	1937.4									
21	1932	4133.12									
22	1933	3616.48									
23	1934	5631.36									
24	1935	5243.88									
25	1936	6328.84									
26	1937	5941.36									
27	1938	7362.12									
28	1939	7103.8									
29	1940	9041.2									
30	1941	9945.32									

Name

Aetna

Anthem

Blue Cross Blue Shield

Cigna Health

Dual Eligible

Humana

Medicaid

Medicare

NO_INSURANCE

UnitedHealthcare

3.5 Dice

Dicing filters the data by selecting specific values from two or more dimensions simultaneously, creating a smaller sub-cube.

Building on the Slice operation, a second slicer was inserted for **Is Covid Flag** from Dim Condition, displaying values: No, Yes, and Unknown.

Two filters were applied simultaneously:

- Payer slicer: **Medicaid** selected
- Is Covid Flag slicer: **Yes** selected

The resulting PivotTable showed only Total Claim Cost for Medicaid encounters that were flagged as COVID-19 related, displaying a Grand Total of **1,420.76** for 2020 only. This demonstrates dicing — the data was filtered across two dimensions (Payer AND Covid Flag) simultaneously to produce a specific sub-cube of data.

1	Row Labels	Total Claim Cost
2	2020	1420.76
3	Grand Total	1420.76
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		
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18		
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21		
22		

Filtered by Payer AND Covid Flag

3.6 Pivot

Pivoting rotates the data axes, swapping rows and columns to view the data from a different perspective without changing the underlying data.

Before pivot — Payers on Rows, Years on Columns: Payer Name was placed in the Rows area and Calendar Hierarchy Year was placed in the Columns area with Total Claim Cost as the value. This showed each payer as a row with years (2015–2020) as columns, making it easy to compare payer performance across years. For example Medicaid showed 81,706.44 in 2015 growing to 387,453.88 in 2020.

After pivot — Years on Rows, Payers on Columns: The axes were swapped — Year was moved to Rows and Payer Name was moved to Columns. The same data is now presented with years as rows and each payer as a column, making it easier to compare all payers within a single year. The Grand Total row confirms the same total of **10,698,341.96** proving the underlying data is unchanged.

This demonstrates pivoting — rotating the view of the data to gain different analytical perspectives without modifying the data itself.

	A	B	C	D	E	F	G	H	I	J
1	Total Claim Cost	Column Labels								
2	Row Labels	2015	2016	2017	2018	2019	2020	Grand Total		
3	Aetna	63650	66336.56	69798	71089.6	238222.56	296990.44	806087.16		
4	Anthem	47143.4	46755.92	46368.44	46626.76	106557	55667.96	349119.48		
5	Blue Cross Blue Shield	81060.64	79071.64	82326.52	91445.2	279682.76	326180.44	939767.2		
6	Cigna Health	68997.24	67783.12	70495.48	75274.4	235613.56	275859.8	794023.6		
7	Dual Eligible	24101.24	25005.28	24876.2	29396.8	73931.12	56003.76	233314.4		
8	Humana	72975.32	73517.84	76617.68	72665.4	248452.08	305825	850053.32		
9	Medicaid	81706.44	87208.56	86640.32	102191.28	389390.96	387453.88	1134591.44		
10	Medicare	292056.56	321350.08	308072.32	339949.04	832048.64	725362.56	2818839.2		
11	NO_INSURANCE	156205.8	164343.04	166538.76	220630.84	648124.4	622111.72	1977954.56		
12	UnitedHealthcare	68842.28	67473.12	68558	74525.16	225383.96	289809.08	794591.6		
13	Grand Total	956738.92	998845.16	1000291.72	1123794.48	3277407.04	3341264.64	10698341.96		
14										
15										

Payers on Rows, Years on Columns

	Total Claim Cost	Column Labels										
2	Row Labels	Aetna	Anthem	Blue Cross Blue Shield	Cigna Health	Dual Eligible	Humana	Medicaid	Medicare	NO_INSURANCE	UnitedHealthcare	Grand Total
3	2015	63650	47143.4	81060.64	68997.24	24101.24	72975.32	81706.44	292056.56	156205.8	68842.28	956738.92
4	2016	66336.56	46755.92	79071.64	67783.12	25005.28	73517.84	87208.56	321350.08	164343.04	67473.12	998845.16
5	2017	69798	46368.44	82326.52	70495.48	24876.2	76617.68	86640.32	308072.32	166538.76	68558	1000291.72
6	2018	71089.6	46626.76	91445.2	75274.4	29396.8	72665.4	102191.28	339949.04	220630.84	74525.16	1123794.48
7	2019	238222.56	106557	279682.76	235613.56	73931.12	248452.08	389390.96	832048.64	648124.4	225383.96	3277407.04
8	2020	296990.44	55667.96	326180.44	275859.8	56003.76	305825	387453.88	725362.56	622111.72	289809.08	3341264.64
9	Grand Total	806087.16	349119.48	939767.2	794023.6	233314.4	850053.32	1134591.44	2818839.2	1977954.56	794591.6	10698341.96
10												
11												
12												
13												

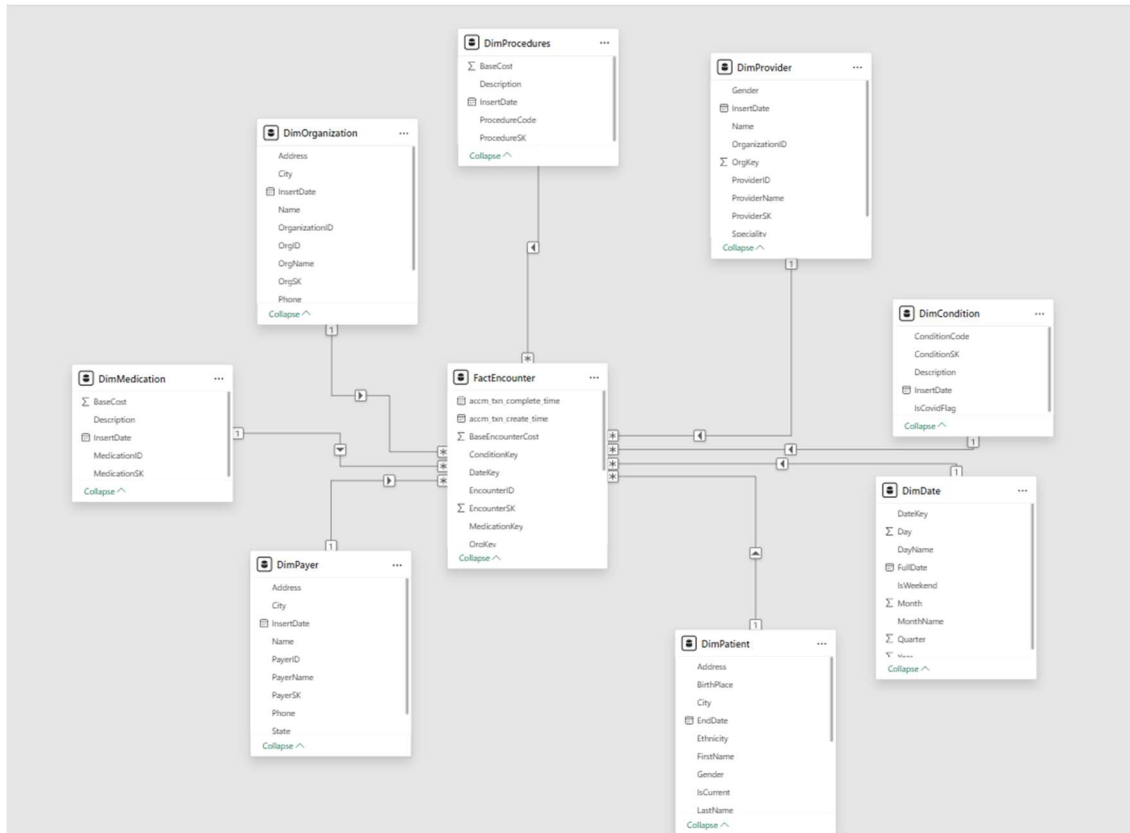
Years on Rows, Payers on Columns

Step 4: PowerBI Reports

4.1 Data Preparation and Modeling

The data preparation and modeling phase established the foundation for all analytical reports:

- Data Connection: Power BI Desktop was connected to the Healthcare_DW data warehouse hosted on the SQL Server instance PC-U510KF5\SQLEXPRESS.
- Modeling: A Star Schema was implemented in the Model view. The FactEncounter table was positioned at the center, surrounded by one-to-many (1:*) relationships with dimension tables like DimDate, DimOrganization, and DimPayer.



- DAX Implementation: To ensure consistent calculations across visuals, an explicit DAX measure was created for financial analysis:

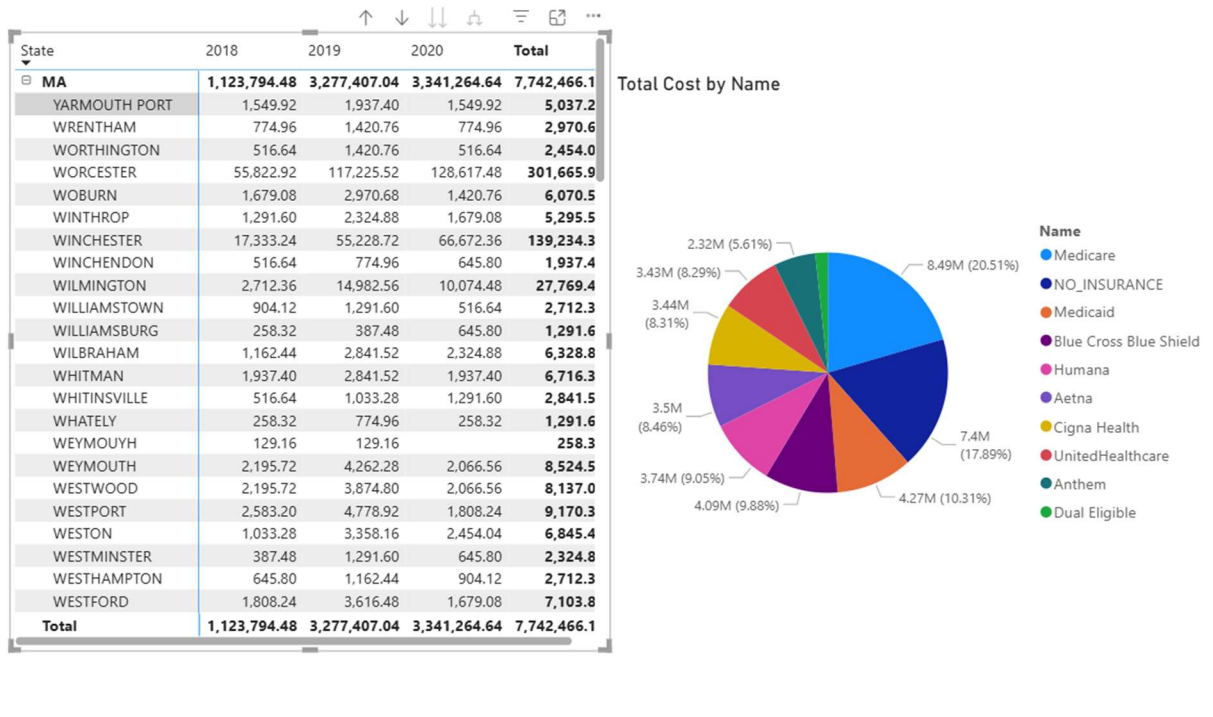
1 Total Cost = SUM(FactEncounter[BaseEncounterCost])

4.2 Report 1: Matrix Visual for Detailed Tabular Data

The first report utilizes a Matrix visual to present a structured, multi-dimensional view of healthcare costs.

- Visual Design: The Matrix visual displays row and column groupings to allow for cross-tabulation analysis.
- Implementation:
 - Rows: State and City from DimOrganization were used as row headers, enabling a geographic hierarchy.
 - Columns: The Year attribute from DimDate was placed in columns to show year-over-year trends.
 - Values: The Total Cost DAX measure provides the aggregated totals at each intersection.

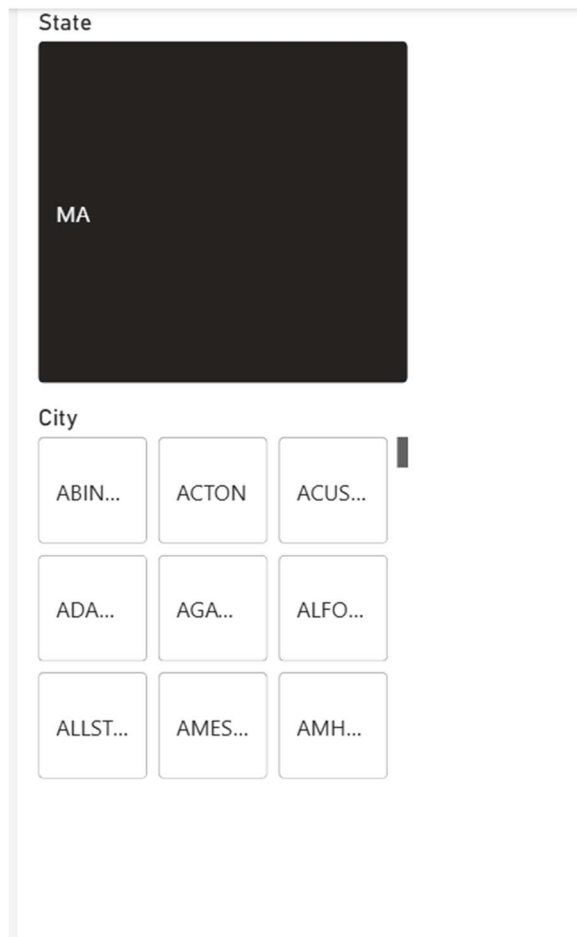
- Insight: This matrix allows users to see the total cost per city (e.g., Acton or Abington) across different fiscal years.



4.3 Report 2: Cascading Filters and Data Insights

This report demonstrates interactive filtering using cascading slicers and multiple visuals to present insights .

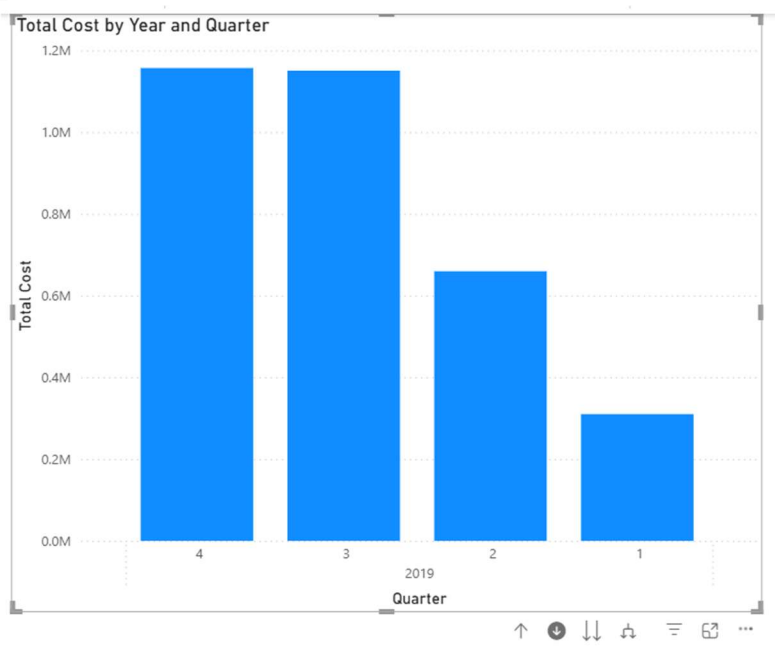
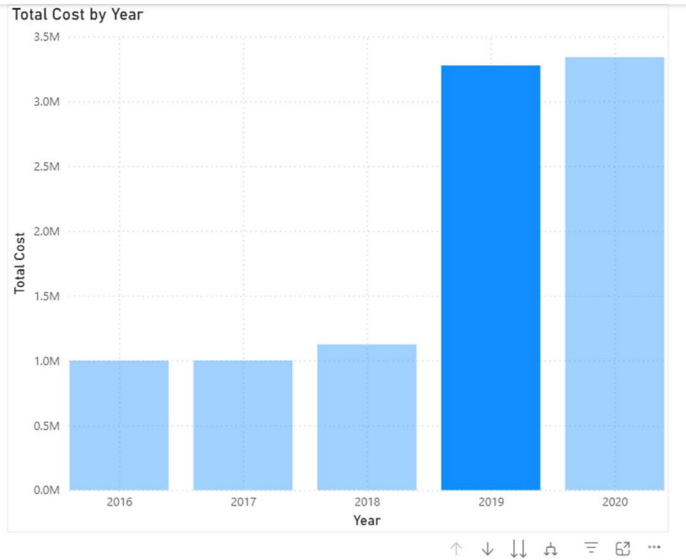
- Cascading Slicers:**
 - Slicer 1 (State): Displays the available states (e.g., MA).
 - Slicer 2 (City): Dynamically updates based on the selection in the State slicer. Because the model uses a star schema, selecting "MA" in the first slicer automatically filters the list of cities in the second.
- Insight Visuals:** A Pie Chart labeled "Total Cost by Name" presents the distribution of costs across different insurance payers, such as Medicare and Medicaid.
- Visual Design:** The combination of slicers and charts ensures that selecting a specific city immediately updates the cost distribution shown in the pie chart.

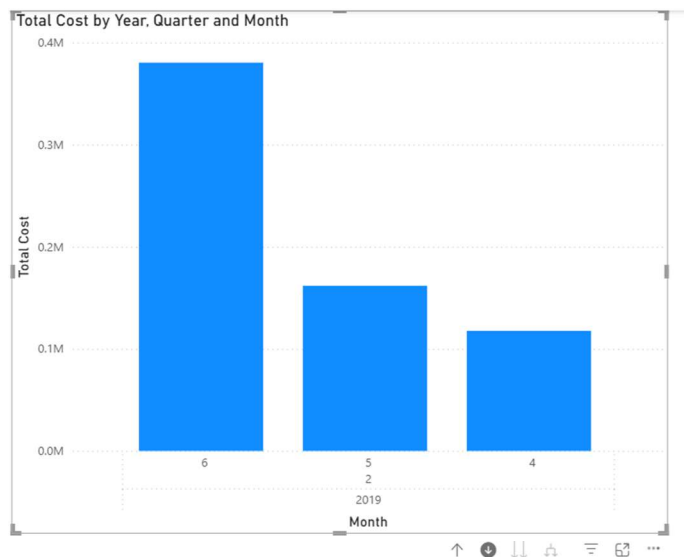


4.4 Report 3: Hierarchical Drill-down Report

This report enables users to explore healthcare costs through different levels of time-based granularity.

- Hierarchy Configuration: A hierarchy was established on the X-axis of a clustered column chart in the following order: Year > Quarter > Month.
- Drill-down Operations:
 - Year Level: Displays total costs for years like 2018, 2019, and 2020.
 - Quarter Level: By enabling the drill-down mode, users can click a year to see its component quarters (Q1, Q2, etc.).
 - Month Level: Further drilling reveals the specific months (e.g., Months 4, 5, 6) within a selected quarter.
- Visual Design: This report provides a seamless way to navigate from broad annual trends down to specific monthly performance.



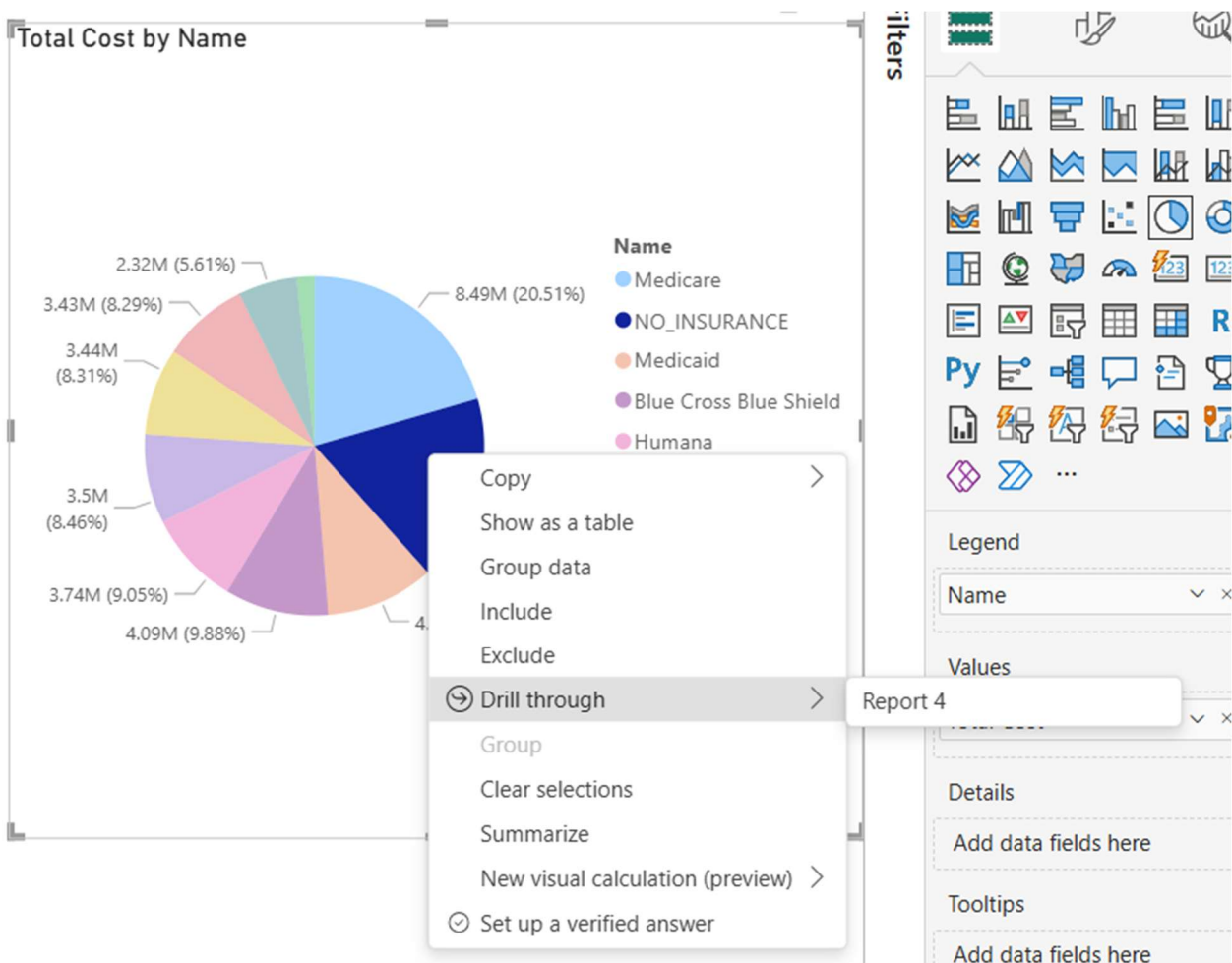


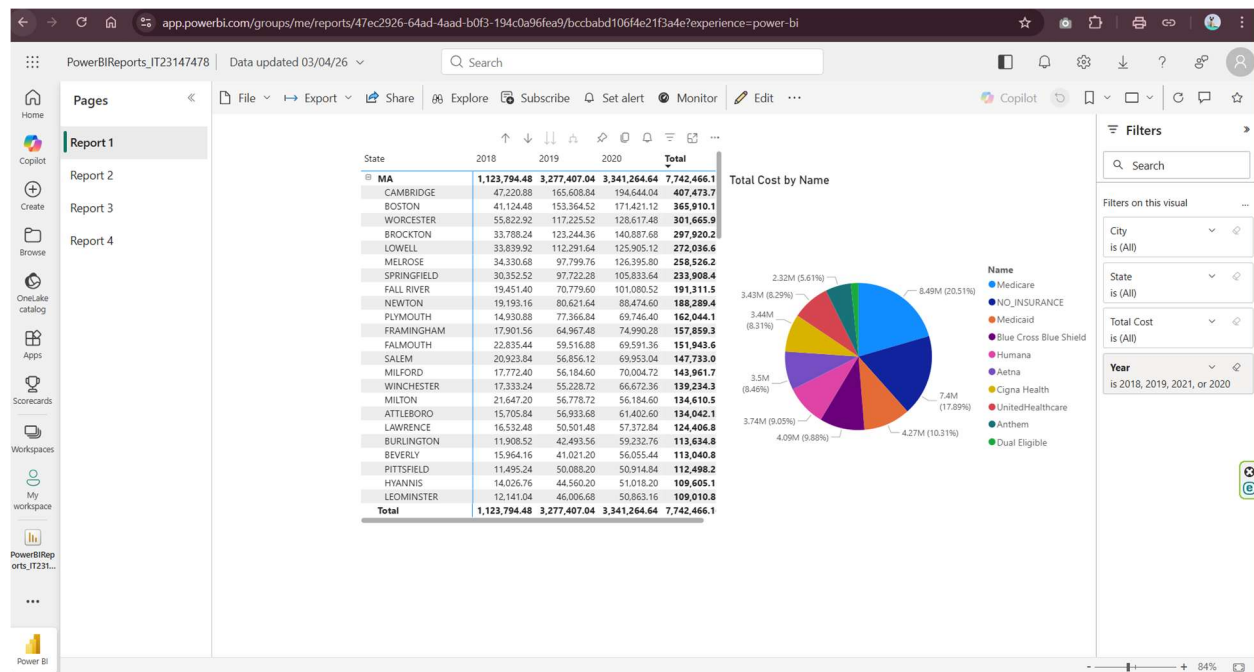
4.5 Report 4: Drill-through for Detailed Navigation

The final report implements a drill-through feature that allows navigation from a summary chart to a highly detailed information page.

- Detail Page (Report 4): A dedicated page was created containing a detailed table visual with columns for FirstName, Condition Description, Medication Description, and Total Cost.
- Drill-through Configuration: The Name field (Payer Name) from DimPayer was dragged into the Drill-through bucket on the detailed page.
- Demonstration:
 - Users can right-click a slice in the "Total Cost by Name" pie chart (e.g., Medicaid).
 - Selecting the Drill through -> Report 4 option navigates the user to the detailed table, automatically filtered for that specific payer.
- Visual Design: This setup bridges the gap between high-level summaries and individual encounter details.

First Name	Condition Description	Medication Description	Total Cost
Abdul218			129.16
Abdul218	Nasal congestion (finding)		129.16
Abe604		Cisplatin 50 MG Injection	1,420.76
Abe604		Verapamil Hydrochloride 40 MG	129.16
Abe604	Cough (finding)		129.16
Abraham100			1,033.28
Abraham100		24 HR Donepezil hydrochloride 10 MG / Memantine hydrochloride 28 MG Extended Release Oral Capsule	129.16
Abraham100	Alzheimer's disease (disorder)	Galantamine 4 MG Oral Tablet	129.16
Abraham100	Headache (finding)		129.16
Abram53			258.32
Abram53		Simvastatin 10 MG Oral Tablet	2,066.56
Abram53	Acute viral pharyngitis (disorder)		129.16
Abram53	Fatigue (finding)		129.16
Abram53	Hyperlipidemia		129.16
Adalberto916		Hydrochlorothiazide 25 MG Oral Tablet	3,099.84
Adalberto916		Simvastatin 10 MG Oral Tablet	3,099.84
Adalberto916	Fever (finding)		129.16
Adalberto916	Viral sinusitis (disorder)		129.16
Adam631			387.48
Adam631		Atenolol 50 MG / Chlorthalidone 25 MG Oral Tablet	2,454.04
Adam631		Donepezil hydrochloride 23 MG Oral Tablet	129.16
Adam631	Acute bronchitis (disorder)	Acetaminophen 21.7 MG/ML / Dextromethorphan Hydrobromide 1 MG/ML / doxylamine succinate 0.417 MG/ML Oral Solution	129.16
Adam631	Alzheimer's disease (disorder)		129.16
Adam631	Neoplasm of prostate	0.25 ML Leuprolide Acetate 30 MG/ML Prefilled Syringe	129.16
Adan632			129.16
Total			8,488,548.48





Published online